

MATLAB Scripts– Sets 16-18

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Set 16:

A toroid whose axis is the z axis carries a current of 5.0 A and has 200 turns. The inner radius is 1.5 cm while the outer radius is 2.5 cm . Write a MATLAB program that computes and plots the magnetic field in the x - y plane in the region $-4.0\text{ cm} \leq x \leq 4.0\text{ cm}$ and $-4.0\text{ cm} \leq y \leq 4.0\text{ cm}$.

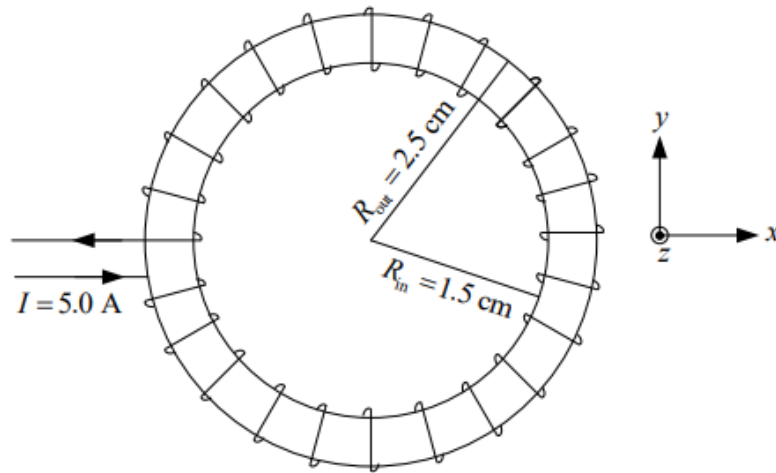


Figure 16.4 The exercise of Set 16.

Analytical Solution:

Given: $R_{in} = 0.015\text{ m}$, $R_{out} = 0.025\text{ m}$, $\mu_0 = 4\pi \times 10^{-7} \frac{\text{H}}{\text{m}}$, $N = 200$ turns, $I = 5.0\text{ A}$,

single – layer toroid centered on the z – axis

Using Ampere's law, for a circular Amperian loop of radius ρ concentric with the toroid axis:

$$\oint \vec{H} \cdot d\vec{L} = I_{enc}$$

By symmetry, $H = H_\phi a_\phi$ and is constant around the loop, so:

$$H_\phi \cdot 2\pi\rho = NI$$

$$\therefore \vec{H} = \frac{NI}{2\pi\rho} a_\phi \text{ for } R_{in} < \rho < R_{out} \text{ and } \vec{H} = 0 \text{ for } \rho < R_{in} \text{ or } \rho > R_{out}$$

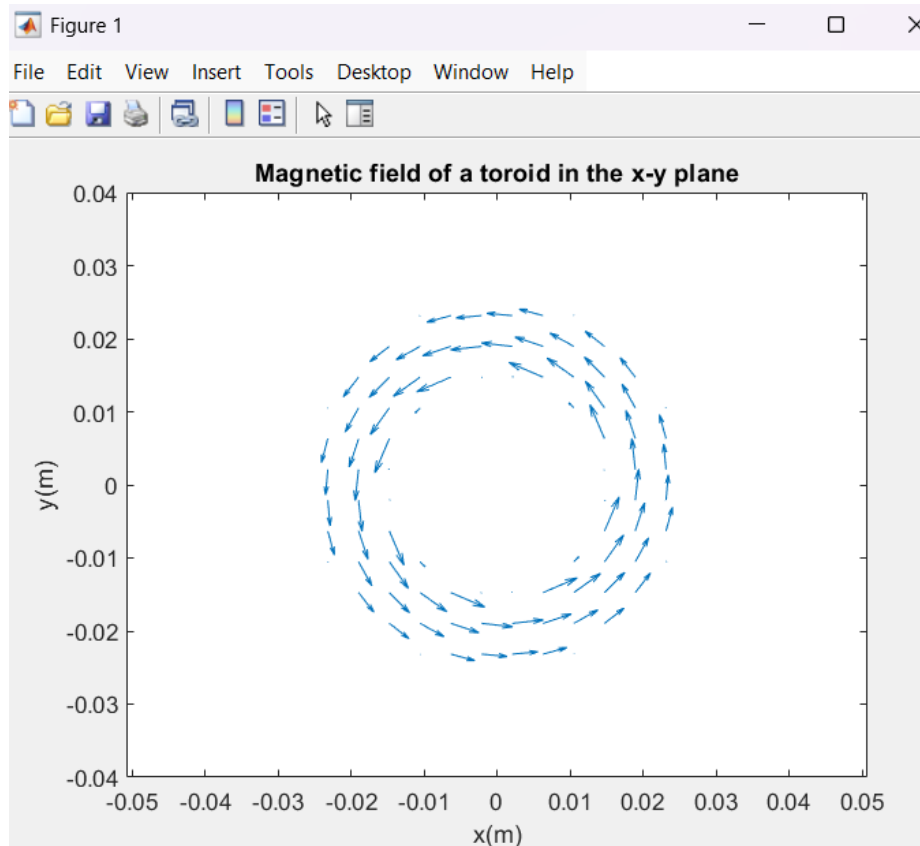
Since this is a toroid with discrete windings, the field varies point – to – point and must be computed numerically using the Biot – Savart Law:

$$\vec{H} = \sum \frac{I \Delta \vec{L}_i \times \vec{R}_i}{4\pi R_i^2}$$

In this formula each segment $\Delta\vec{L}_i$ of the toroidal winding is parameterized as a helix wrapped around the toroidal core. The MATLAB program constructs the winding geometry, discretizes it into small current elements, and computes $\vec{H}$ at each point on the $x - y$ plotting grid via superposition.

$\therefore$ the analytical result from Ampere's law serves as a benchmark to verify the MATLAB numerical solution. The plot below confirms the analytical result from Ampere's law. The magnetic field is concentrated within the toroid in the ϕ direction and is approximately zero outside, consistent with $H = \frac{NI}{2\pi\rho} a_\phi$. The numerical Biot – Savart solution could be made more accurate by increasing the number of segments (currently 20,000) and using a finer observation grid (currently 20x20), which would reduce discretization error at the cost of greater computational expense.

MATLAB Output/Plot:



Set 17:

Two coils with radii a_1 and a_2 are separated by a distance of d as shown in Figure 17.3. The dimensions are $a_1 = 0.01\text{ m}$, $a_2 = 0.04\text{ m}$, and $d = 0.1\text{ m}$. Find the mutual inductance of the coils. Write a MATLAB program to verify your answer.

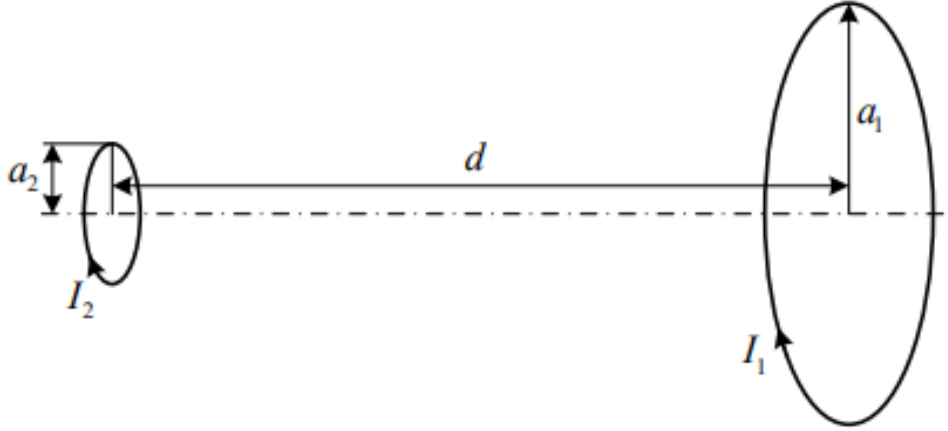


Figure 17.3 The geometry of the exercise of Set 17.

Analytical Solution:

Given: $a_1 = 0.01\text{ m}$, $\mu_0 = 4\pi \times 10^{-7} \frac{\text{H}}{\text{m}}$, $a_2 = 0.04\text{ m}$, $d = 0.1\text{ m}$, singleturn coils, coaxial

The B – field on the axis of a circular loop of radius a_2 carrying current I_2 , at a distance d from the center, is:

$$B_z = \frac{\mu_0 I_2 a_2^2}{2(a_2^2 + d^2)^{\frac{3}{2}}}$$

Since a_1 is small compared to d , we can assume B is approximately uniform over the area of coil 1. The flux through coil 1 is:

$$\Phi_{12} = B_z \cdot \pi a_1^2$$

$$\Phi_{12} = \frac{\mu_0 \pi I_2 a_1^2 a_2^2}{2(a_2^2 + d^2)^{\frac{3}{2}}}$$

$\therefore$ the mutual inductance is:

$$M = \frac{\Phi_{12}}{I_2} = \frac{\frac{\mu_0 \pi I_2 a_1^2 a_2^2}{2(a_2^2 + d^2)^{\frac{3}{2}}}}{I_2} = \frac{\mu_0 \pi a_1^2 a_2^2}{2(a_2^2 + d^2)^{\frac{3}{2}}}$$

Sub in values:

$$M = \frac{\left(4\pi \times 10^{-7} \frac{H}{m}\right) \pi (0.01 \text{ m})^2 (0.04 \text{ m})^2}{2[(0.04 \text{ m})^2 + (0.1 \text{ m})^2]^{\frac{3}{2}}}$$
$$M \approx 2.528 \times 10^{-10} \text{ H}$$

$\therefore$ the analytical mutual inductance is approximately $2.528 \times 10^{-10} \text{ H}$

$\therefore$ the mutual inductance between the two coaxial single – turn coils separated by a distance of 0.1 m is approximately $2.528 \times 10^{-10} \text{ H}$ analytically. This was close to the result the MATLAB code generated verifying our results. The MATLAB code generated an answer of $\approx 2.4995 \times 10^{-10} \text{ H}$ for the mutual inductance which only has a 1.13% error to the analytical result. This small discrepancy arises because the analytical solution assumed the magnetic field is approximately uniform over the area of coil 1, using the on – axis B – field value evaluated at the center of the coil, which is a valid approximation since a_1 is small compared to d . The MATLAB code, however, does not make this assumption and instead computes the magnetic field at every surface element across coil 1, capturing the slight variation in field strength away from the axis. As a result, the MATLAB answer is marginally lower than the analytical answer and represents the more accurate value. To generate an even closer result, the number of discretization steps for the integration in the MATLAB code would be increased (equivalently, the step size would be decreased), which would improve accuracy at the cost of greater computational expense.

MATLAB Output to Verify Answers:

```
Command Window
The mutual inductance of the coils separated by a distance of 0.1m is: 2.499467e-10 H
fx >> |
```

Set 18:

A toroid whose axis is the z axis has **200 turns**. The inner radius is **2.0 cm** while the outer radius is **2.5 cm**. Find the inductance analytically if the toroid has 200 turns and $\mu_r = 1$. Write a MATLAB program to verify your answer.

Analytical Solution:

$$\text{Given: } N = 200 \text{ turns, } R_{in} = 0.02 \text{ m, } R_{out} = 0.025 \text{ m, } \mu_r = 1, \mu_0 = 4\pi \times 10^{-7} \frac{H}{m}$$

Geometry (Assume circular cross section as no height given for rectangular cross section):

$$R = \frac{R_{in} + R_{out}}{2} = \frac{0.02 \text{ m} + 0.025 \text{ m}}{2} = 0.0225 \text{ m}$$

$$a = \frac{R_{out} - R_{in}}{2} = \frac{0.025 \text{ m} - 0.02 \text{ m}}{2} = 0.0025 \text{ m}$$

Formula for circular cross – section toroid ($a \ll R$):

$$L = \frac{\mu_0 N^2 \pi a^2}{2\pi R} = \frac{\mu_0 N^2 a^2}{2R}$$

Sub in values:

$$L = \frac{\left(4\pi \times 10^{-7} \frac{H}{m}\right) (200)^2 (0.0025 \text{ m})^2}{2(0.0225 \text{ m})}$$

$$L \approx 6.981 \times 10^{-6} \text{ H}$$

$\therefore$ the analytical inductance for the toroid is approximately $6.981 \times 10^{-6} \text{ H}$

$\therefore$ the self – inductance of the toroid with 200 turns, inner radius 2.0 cm, and outer radius 2.5 cm is approximately $6.981 \times 10^{-6} \text{ H}$ analytically. This was close to the result the MATLAB code generated verifying our results. The MATLAB code generated an answer of $\approx 7.380 \times 10^{-6} \text{ H}$ for the self – inductance which only has a 5.72% error to the analytical result. This discrepancy arises because the analytical solution uses the thin – ring approximation, which assumes the magnetic field is uniform across the toroid's circular cross – section and evaluated at the mean radius R , a valid approximation when the cross – sectional radius a is small compared to R . In

this case, $\frac{a}{R} = \frac{1}{9}$ is not extremely small, so the field varies noticeably across the

cross – section, being stronger on the inner side where $\rho < R$. The MATLAB code, however, does not make this assumption and instead models the winding as a continuous helical path and computes the magnetic field at every surface element across the cross – section, capturing the field variation with radial position. As a result, the MATLAB answer is marginally higher than the analytical answer and represents the more accurate value. To generate an even closer result, the number of discretization steps for the winding segments and surface elements in the MATLAB code would be increased (equivalently, the step size would be decreased), which would improve accuracy at the cost of greater computational expense.

MATLAB Output to Verify Answers:

Command Window

```
The self-inductance of the toroid is: 7.380033e-06 H  
fx >>
```